

Repair 1 - Environment and Python basics

Scope: PY0-PY1

- Print interpreter/version/path before reinstalling packages.
- Restart + Run All to expose hidden state.
- Test business-rule boundaries just below/exactly/above.
- Prefer small readable functions over clever code.

Repair discipline

- Keep the failed notebook/evidence.
- Repair the earliest weak stage.
- Close review before changed retry.
- Restart + Run All after an upstream repair.

Repair 2 - Load, clean and aggregate

Scope: PY2-PY4

- Inspect shape/dtypes/missing/key evidence before analysis.
- Detect before correcting; keep an issue log.
- Never use blanket drop/fill without business rule.
- State grouped result grain and reconcile additive totals.

Repair discipline

- Keep the failed notebook/evidence.
- Repair the earliest weak stage.
- Close review before changed retry.
- Restart + Run All after an upstream repair.

Repair 3 - Merge and reshape

Scope: PY5-PY6

- Clean/prove dimension keys before merge.
- Keep `validate=` and `indicator=True`; investigate `\_merge`.
- Do not repair a failed uniqueness assertion by disabling it.
- Reshape long/wide/long and reconcile row meaning/totals.

Repair discipline

- Keep the failed notebook/evidence.
- Repair the earliest weak stage.
- Close review before changed retry.
- Restart + Run All after an upstream repair.

Repair 4 - Visualization, EDA and reproducibility

Scope: PY7-PY9

- Validate summary before plotting.
- Create separate purposeful figures with labels.
- Separate data-quality findings from business observations; correlation is not causation.
- Use relative output paths and pass Restart Kernel -> Run All.

Repair discipline

- Keep the failed notebook/evidence.
- Repair the earliest weak stage.
- Close review before changed retry.
- Restart + Run All after an upstream repair.

Repair 5 - Integration and Gate

Scope: PY10 / M-PY / Gate

- Use the smallest useful workflow.
- Bring sales/returns to compatible grain before combining money.
- Reconcile critical counts/totals before recommendations.
- Gate B/C require prior failure + saved attempt + review + targeted repair + explicit assignment; no automatic progression.

Repair discipline

- Keep the failed notebook/evidence.
- Repair the earliest weak stage.
- Close review before changed retry.
- Restart + Run All after an upstream repair.